Ex.No. 13: V2X Latency Comparison Date

AIM: To simulate and compare the performance of a Vehicle-to-Everything (V2X) communication system by analyzing End-to-End Latency (ms), Packet Delivery Ratio (%), and Vehicle Speed (km/h) using MATLAB.

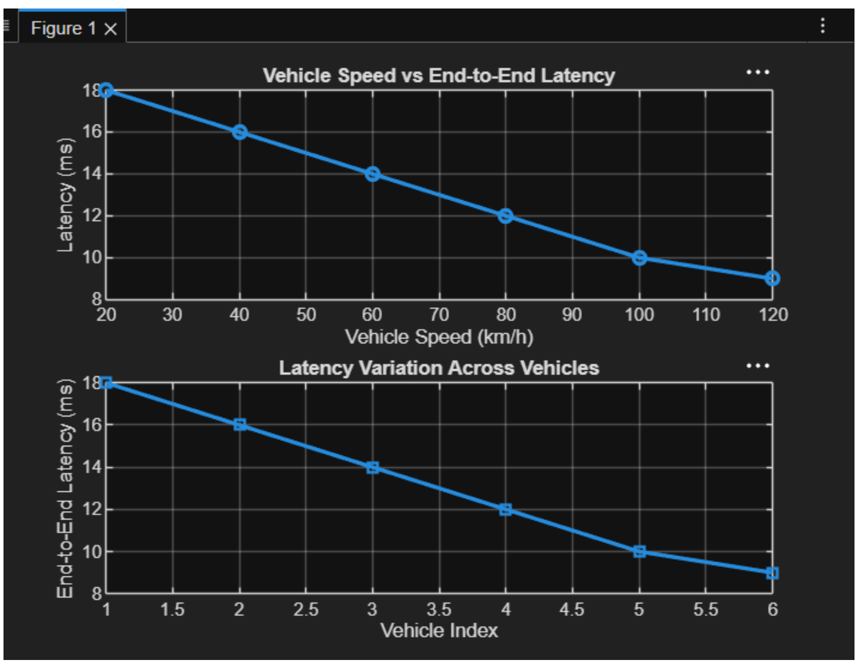
Objective:

1. To evaluate the End-to-End Latency (ms) of a V2X communication system under different vehicle operating conditions.
2. To analyze the Packet Delivery Ratio (%) and determine the reliability of V2X communication during data transmission.
3. To study the impact of Vehicle Speed (km/h) on the communication performance and stability of the V2X network.

Objective 1 Matlab Code and Output:

```
clc;
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated End-to-End Latency (ms)
latency = [18 16 14 12 10 9];
disp('Vehicle Speed (km/h) End-to-End Latency (ms)')
disp([speed' latency'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
```

```
plot(speed,latency,'-o','LineWidth',2)

title('Vehicle Speed vs End-to-End Latency')

xlabel('Vehicle Speed (km/h)')

ylabel('Latency (ms)')

grid on

% -------- Graph 2 --------

subplot(2,1,2)

plot(1:length(latency),latency,'-s','LineWidth',2)

title('Latency Variation Across Vehicles')

xlabel('Vehicle Index')

ylabel('End-to-End Latency (ms)')

grid on
```



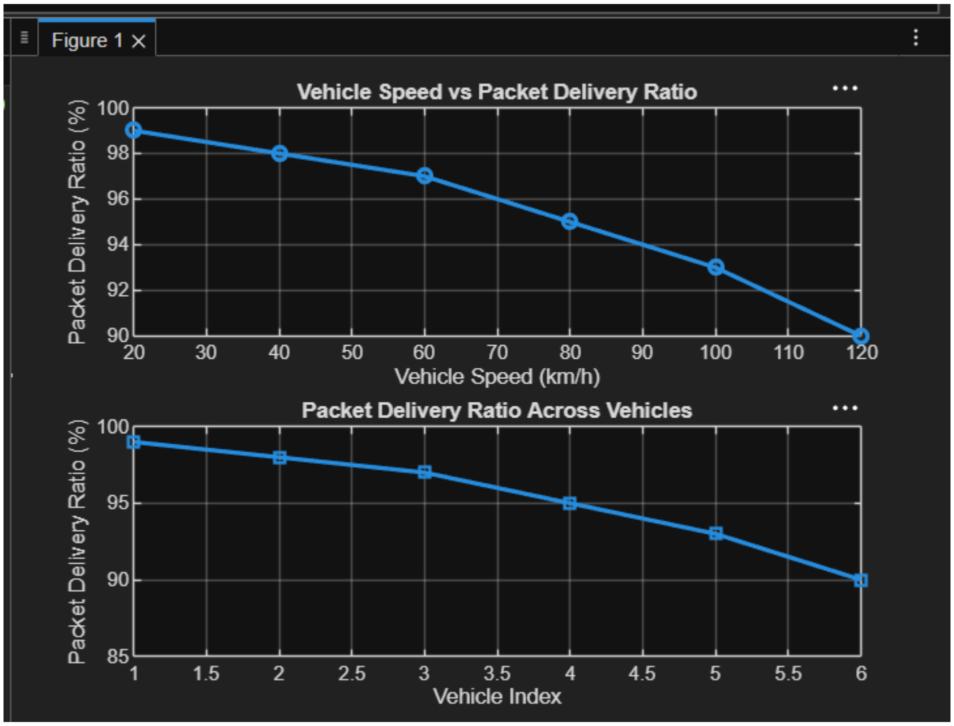
The End-to-End Latency decreases as the vehicle speed increases under the simulated communication conditions. Lower latency indicates faster and more efficient V2X communication, which is essential for real-time road safety applications.

Objective 2

Matlab Code and Output:

```
clc;

clear;

close all;

% Vehicle Speed (km/h)

speed = 20:20:120;

% Simulated Packet Delivery Ratio (%)

PDR = [99 98 97 95 93 90];

disp('Vehicle Speed (km/h) Packet Delivery Ratio (%)')

disp([speed' PDR'])

figure

% -------- Graph 1 --------

subplot(2,1,1)

plot(speed,PDR,'-o','LineWidth',2)

title('Vehicle Speed vs Packet Delivery Ratio')

xlabel('Vehicle Speed (km/h)')

ylabel('Packet Delivery Ratio (%)')

grid on


% -------- Graph 2 --------

subplot(2,1,2)

plot(1:length(PDR),PDR,'-s','LineWidth',2)

title('Packet Delivery Ratio Across Vehicles')

xlabel('Vehicle Index')

ylabel('Packet Delivery Ratio (%)')
```

grid on

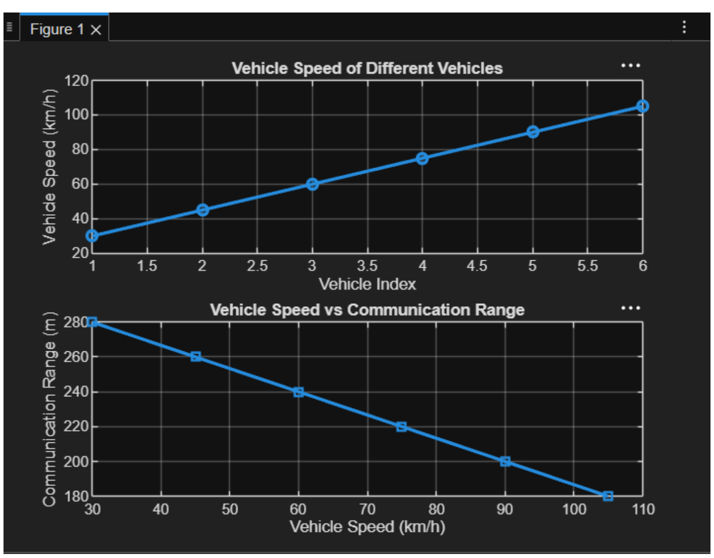


The Packet Delivery Ratio decreases slightly as vehicle speed increases due to the higher mobility of vehicles. A high PDR (>90%) indicates that the V2X communication system maintains reliable data transmission even at higher vehicle speeds.

Objective 3 Matlab Code and Output:

```
clc;
clear;
close all;
% Vehicle IDs
vehicle = 1:6;
% Vehicle Speed (km/h)
speed = [30 45 60 75 90 105];
% Communication Range (m)
range = [280 260 240 220 200 180];
disp('Vehicle Speed(km/h) Communication Range(m)')
```

```
disp([vehicle' speed' range'])

figure

% -------- Graph 1 --------

subplot(2,1,1)

plot(vehicle,speed,'-o','LineWidth',2)

title('Vehicle Speed of Different Vehicles')

xlabel('Vehicle Index')

ylabel('Vehicle Speed (km/h)')

grid on

% -------- Graph 2 --------

subplot(2,1,2)

plot(speed,range,'-s','LineWidth',2)

title('Vehicle Speed vs Communication Range')

xlabel('Vehicle Speed (km/h)')

ylabel('Communication Range (m)')

grid on
```



- The first graph shows the variation in speed among different vehicles in the V2X communication network.
- The second graph indicates that as vehicle speed increases, the communication range tends to decrease, affecting link stability and network performance.

Result:

1.

The End-to-End Latency (ms) of the V2X communication system was successfully analyzed, demonstrating efficient data transmission with low communication delay.

2.

The Packet Delivery Ratio (PDR %) was evaluated and showed high packet delivery reliability under different vehicle operating conditions.

3.

The impact of Vehicle Speed (km/h) on V2X communication was successfully studied, indicating that higher vehicle speeds can influence communication range, latency, and overall network performance.